



Care-Gap Prediction for Chronic Disease Management

Identifying patients likely to miss recommended screenings or follow-ups, to close gaps before they become costly complications | Sohail — Strategic HealthCare BI Analyst

15,000

Chronic Disease
Patients Analyzed

0.716

Best ROC-AUC
(Logistic Regression)

70%

Precision @ Top 10%
(vs. 34.6% base rate)

6.40x

ROI at 5% Capacity Tier

\$878K

Est. Annual Net Savings
(50K-patient panel)

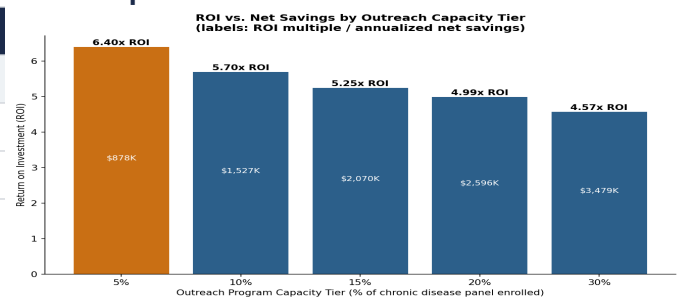
Headline finding: An interpretable logistic regression model — trained on utilization, adherence, and access features rather than clinical severity alone — outperformed both tree ensembles at identifying which chronic disease patients will miss their next guideline-recommended screening or follow-up. Days since last visit, appointment no-show history, and reminder staleness dominate the risk signal, meaning most of the gap is closeable through outreach timing and channel — not a clinical severity problem.

Model Comparison (Held-Out Test Set)

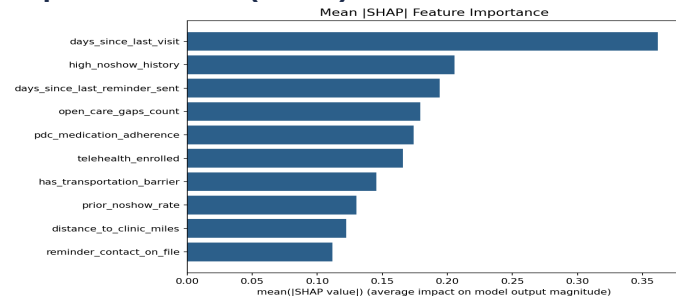
MODEL	ROC-AUC	PR-AUC	PREC @ TOP 10%
Logistic Regression BEST	0.716	0.583	70.0%
Random Forest	0.711	0.569	66.0%
XGBoost	0.707	0.567	68.3%

The interpretable baseline edged out both tree ensembles — consistent with the readmission project's finding that, at this feature-set size, linear models are competitive and easier to trust operationally.

Cost / Impact Simulation



Top Risk Drivers (SHAP)



- **Days since last visit** and **reminder staleness** are the two strongest drivers — both are operational levers, not clinical ones.
- **Prior no-show history** and **medication adherence (PDC)** confirm the model is picking up genuine behavioral risk, not just demographics.
- **Telehealth enrollment** and having a **reminder contact on file** both push risk down — concrete, actionable enrollment targets for the outreach team.
- ROI peaks at the narrowest 5% tier (6.40x) but total annual net savings keep climbing through 30% capacity — the same staffing tradeoff seen in the readmission project.

Key Modeling Decisions

Why precision@K over ROC-AUC alone

An outreach team can only call a fixed share of the panel each cycle. Precision at the 10–20% capacity threshold reflects what matters operationally: how often the flagged list is actually right, not how the model ranks the full population.

Why the target label matches the guideline window exactly

"Missed next service" is defined against each condition's specific guideline window (e.g., annual eye exam, semi-annual HbA1c), not a generic 30/60/90-day cutoff — so the label means the same thing a care-gap registry would flag.

Why class weighting over resampling

Imbalance (34.6% positive rate) was handled via `class_weight="balanced"` rather than synthetic oversampling, to keep predicted probabilities calibrated for risk-tiering.

Honest Limitations & Next Steps

Synthetic data, real-shaped signal. Risk-factor directions (no-show history, adherence, reminder timing) reflect published care-gap literature, but coefficients are illustrative, not fit to a real population.

Before production: validate against a real care-gap registry with a temporal holdout, confirm the 35% relative-risk-reduction assumption against pilot data, and audit for disparate impact across insurance type and language access before setting enrollment thresholds.

Stack: Python · pandas · scikit-learn · XGBoost · SHAP | **Data:** Synthetic chronic-disease population (diabetes, hypertension, CHF, CKD, COPD) | **Eval:** ROC-AUC, PR-AUC, precision@K, cost/ROI simulation